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PROJECT ON

# World Wide Energy Consumption Analysis


- By
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# About me

- **Background? (B-tech or M-tech)**

  I have completed my Bachelor of Technology (B.Tech) in CSE from BVRIT Hyderabad College of Engineering for Women, Hyderabad, as part of the 2021–2025 batch.
- **Why you want to learn Data Science**

  I want to learn Data Science because I am interested in working with data and understanding how it can be used to solve real-world problems. It combines programming, analytical thinking, and problem-solving, which are skills I enjoy. I believe learning Data Science will help me build a strong career and work on impactful projects.
- **Any work experience**

  Software Developer Intern, S&P Global (Third Year B.Tech)
  Software Engineer, Capgemini (After Graduation)
- **Share your LinkedIn and GitHub profile URLs**

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# Agenda

- Introduction
- Objective of the Project
- Key Analysis Questions
- Conclusion
- Experience/Challenges
- Questions & Answers

# Introduction

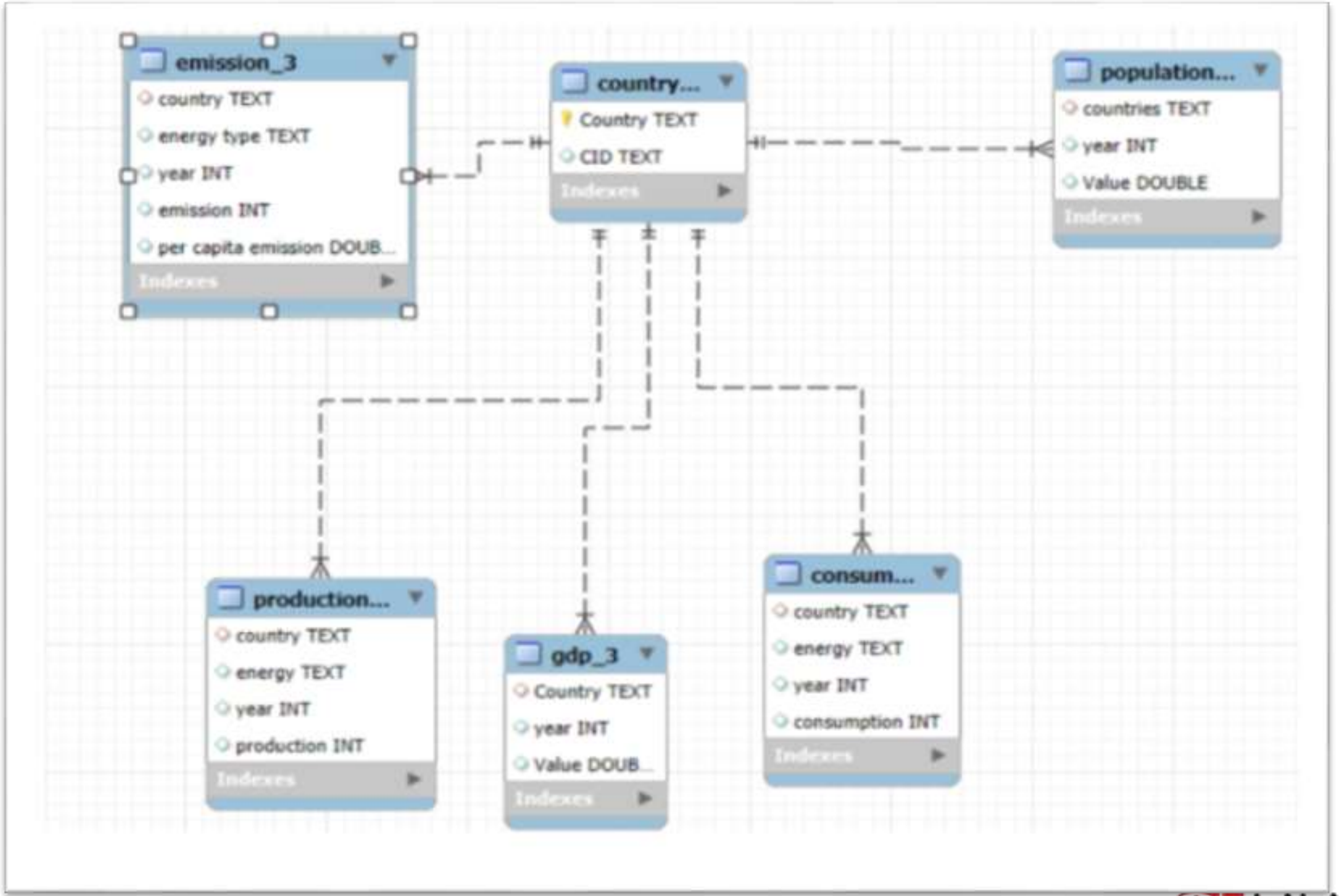
- Energy is one of the key drivers of economic growth and development across the world.
- Countries differ in their energy consumption, energy production, emissions, GDP, and population patterns.
- Analyzing global energy data helps in understanding consumption trends, environmental impact, and economic performance.
- This project focuses on studying worldwide energy consumption and its relationship with emissions, GDP, and population across different countries and years.
- The analysis aims to uncover meaningful insights and identify patterns that reflect real-world energy and economic conditions.

## Objective of the Project

- The main objective of this project is to analyze and understand the relationship between a country's energy consumption, energy production, emissions, GDP, and population.

- Using SQL, the project explores real-world data from different countries to identify trends, compare performance, and gain meaningful insights into how economic growth and energy usage impact the environment.

# ER Diagram:

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# Key analysis questions (use cases)

## Use Case 1: *Total emission per country for the most recent year available?*

```
select Country,sum(emission) as total_emission
from emission
where year=(select max(year) from emission)
group by Country;
```

| Country | total_emission |
|---|---|
| Afghanistan | 16 |
| Albania | 7 |
| Algeria | 328 |
| American Samoa | 0 |
| Angola | 39 |
| Antarctica | 0 |
| Antigua and Barbuda | 2 |
| Argentina | 396 |
| Armenia | 14 |
| Aruba | 2 |
| Australia | 789 |
| Austria | 114 |
| Azerbaijan | 78 |
| Bahrain | 96 |
| Bangladesh | 252 |
| Barbados | 2 |
| Belarus | 93 |
| Belgium | 224 |
| Belize | 2 |
| Benin | 11 |
| Bermuda | 2 |
| Bhutan | 1 |
| Bolivia | 44 |
| Bosnia and Herzeg... | 49 |
| Botswana | 12 |
| Brazil | 876 |
| British Virgin Islands | 0 |
| Brunei | 22 |

## Observation:

The analysis identifies the countries contributing the highest emissions in the latest available year.

**Use Case 2:** *Top 5 countries by GDP in the most recent year?*

select Country,gdp_value
from gdp
where year=(select max(year) from gdp)
order by gdp_value desc limit 5;

| Country | gdp_value |
|---|---|
| China | 28673.24 |
| United States | 22679.47 |
| India | 11660.21 |
| Japan | 5179.704 |
| Germany | 4463.949 |

## Observation:

The results show the world's largest economies based on GDP.

## **Use Case 3:** *Compare energy production and consumption by country and year.*

select production.Country, production.year, sum(production.production) as total_production,sum(consumption.consumption) as total_consumption

from production

inner join consumption

on production.Country=consumption.Country and production.year=consumption.year

group by production.Country,production.year

Order by Country,year;

| Country | year | total_producti... | total_consumpti... |
|---|---|---|---|
| Afghanistan | 2020 | 0 | 0 |
| Afghanistan | 2021 | 0 | 0 |
| Afghanistan | 2022 | 0 | 0 |
| Afghanistan | 2023 | 0 | 0 |
| Albania | 2020 | 0 | 0 |
| Albania | 2021 | 0 | 0 |
| Albania | 2022 | 0 | 0 |
| Albania | 2023 | 0 | 0 |
| Algeria | 2020 | 36 | 18 |
| Algeria | 2021 | 42 | 18 |
| Algeria | 2022 | 42 | 18 |
| Algeria | 2023 | 42 | 18 |
| American... | 2020 | 0 | 0 |
| American... | 2021 | 0 | 0 |
| American... | 2022 | 0 | 0 |
| American... | 2023 | 0 | 0 |

## **Observation:**

The comparison reveals whether countries produce enough energy to meet their demand or rely on external sources. Some countries show higher production than consumption, while others consume more than they produce.

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## **Use Case 4:** *Which Energy types contribute most to emissions across all countries?*

```
select energytype,sum(emission) as total
from emission
group by energytype
order by total desc;
```

| energytype | total |
|---|---|
| CO2 emissions (MMtonnes CO2) | 142723 |
| Coal and coke (MMtonnes CO2) | 63945 |
| Petroleum and other liquids (MMtonnes CO2) | 47297 |
| Consumed natural gas (MMtonnes CO2) | 31469 |

## **Observation:**

The analysis identifies which energy sources contribute the most to overall emissions. This helps understand the environmental impact of different energy types.

**<u>Use Case 5:</u>** *How have global emissions changed year over year?*

select year,sum(emission) as total_emissions

from emission

group by year

order by year;

| year | total_emissio... |
|---|---|
| 2020 | 67852 |
| 2021 | 70976 |
| 2022 | 72445 |
| 2023 | 74161 |

## Observation:

The year-wise emission trend shows whether global emissions are increasing, decreasing, or remaining stable over time.

## **Use Case 6:** *The trend in GDP for each country over the give years?*

```
select Country,year,gdp_value
from gdp
order by Country,year;
```

| Country | year | gdp_value |
|---|---|---|
| Afghanistan | 2020 | 83.21645 |
| Afghanistan | 2021 | 65.95827 |
| Afghanistan | 2022 | 61.84236 |
| Afghanistan | 2023 | 58.90645 |
| Afghanistan | 2024 | 57.83112 |
| Albania | 2020 | 36.78752 |
| Albania | 2021 | 40.04762 |
| Albania | 2022 | 41.99279 |
| Albania | 2023 | 43.63172 |
| Albania | 2024 | 45.2091 |
| Algeria | 2020 | 531.9749 |
| Algeria | 2021 | 552.5308 |
| Algeria | 2022 | 572.6695 |
| Algeria | 2023 | 596.0418 |
| Algeria | 2024 | 615.7547 |

## **Observation:**

The GDP trend analysis helps track the economic growth of countries over multiple years. It highlights countries with consistent growth, fluctuations of economic slowdown.

**Use Case 7:** *Has energy consumption increased or decreased over the years for major economies?*

```
select Country,year,sum(consumption)
from consumption
group by Country,year
order by Country,year;
```

| Country | year | sum(consumpti... |
|---|---|---|
| Afghanistan | 2020 | 0 |
| Afghanistan | 2021 | 0 |
| Afghanistan | 2022 | 0 |
| Afghanistan | 2023 | 0 |
| Albania | 2020 | 0 |
| Albania | 2021 | 0 |
| Albania | 2022 | 0 |
| Albania | 2023 | 0 |
| Algeria | 2020 | 3 |
| Algeria | 2021 | 3 |
| Algeria | 2022 | 3 |
| Algeria | 2023 | 3 |
| American... | 2020 | 0 |
| American... | 2021 | 0 |
| American... | 2022 | 0 |

## Observation:

The results indicate how energy demand has changed over time. An increasing trend suggests growing industrial and economic activities, while a decreasing trend may indicate improved efficiency or changes in economic conditions.

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## <u>**Use Case 8:**</u> *What is the emission-to-GDP ratio for each country by year?*

```
select emission.Country,sum(emission.emission)/sum(gdp.gdp_value) as ratio,emission.year
from emission
inner join gdp
on emission.Country=gdp.Country and emission.year=gdp.year
group by Country,year
order by Country,year;
```

| Country | ratio | year |
|---|---|---|
| Afghanistan | 0.05407584678269742 | 2020 |
| Afghanistan | 0.10107400734838355 | 2021 |
| Afghanistan | 0.09702087695230259 | 2022 |
| Afghanistan | 0.09053903831131113 | 2023 |
| Albania | 0.04077469750611077 | 2020 |
| Albania | 0.04994054578024861 | 2021 |
| Albania | 0.0535806265789913 | 2022 |
| Albania | 0.040108434872611023 | 2023 |
| Algeria | 0.1334649435527879 | 2020 |
| Algeria | 0.13754889320197172 | 2021 |
| Algeria | 0.13838697538458047 | 2022 |
| Algeria | 0.13757424395403142 | 2023 |
| Angola | 0.033580112421584644 | 2020 |
| Angola | 0.03893906174269251 | 2021 |

## <u>**Observation:**</u>

This ratio measures the amount of emissions generated per unit of economic output.

## <u>Use Case 9:</u> *Which countries have the highest energy consumption relative to GDP?*

```
select consumption.Country,sum(consumption.consumption/gdp.gdp_value) as ratio
from consumption
inner join gdp
on consumption.Country=gdp.Country
group by Country
order by ratio desc;
```

| Country | ratio |
|---|---|
| Trinidad and Tobago | 0.605434936651486 |
| North Korea | 0.534304796361005 |
| Turkmenistan | 0.400925467613196 |
| Bahrain | 0.255393447377952 |
| Kuwait | 0.2058568422565602 |
| Iran | 0.1991135946215989 |
| Qatar | 0.1773518011808067 |
| Ukraine | 0.1769357730034369 |
| Russia | 0.173871439906097 |
| Kazakhstan | 0.1602997681624645 |
| Iraq | 0.1561141979254549 |
| Canada | 0.1460460075198874 |
| United Arab Emirates | 0.137397969017425 |
| Venezuela | 0.13352986583723 |
| Saudi Arabia | 0.130630116286480 |
| Oman | 0.1294113893983990 |
| South Africa | 0.1287413915839587 |
| China | 0.1286088068030293 |
| Singapore | 0.118682241542180 |
| South Korea | 0.1123206001629463 |
| Belarus | 0.1118200641954525 |
| Algeria | 0.1048512924192748 |

## <u>Observation:</u>

The analysis identifies countries that consume large amounts of energy compared to their economic output.

## Use Case 10: *What is the global share (%) of emissions by country?*

```
select Country,sum(emission) as country_emission,sum(emission)/(select sum(emission) from emission) * 100 as global_percentage

from emission

group by Country;
```

| Country | country_emissi... | global_percenta... |
|---|---|---|
| Afghanistan | 72 | 0.0300 |
| Albania | 30 | 0.0100 |
| Algeria | 1233 | 0.4300 |
| American Samoa | 0 | 0.0000 |
| Angola | 141 | 0.0500 |
| Antarctica | 0 | 0.0000 |
| Antigua and Barbuda | 8 | 0.0000 |
| Argentina | 1534 | 0.5400 |
| Armenia | 58 | 0.0200 |
| Aruba | 8 | 0.0000 |
| Australia | 3110 | 1.0900 |
| Austria | 468 | 0.1600 |
| Azerbaijan | 303 | 0.1100 |
| Bahrain | 358 | 0.1300 |

## Observation:

The results show each country's contribution to total global emissions.

## Conclusion

- This project helped me to analyze the relationship between energy consumption, energy production, emissions, GDP, and population across different countries.
- The analysis showed that economic growth and energy consumption are closely related, while emissions vary depending on energy usage patterns and efficiency.
- The project also highlighted differences in per-capita emissions, energy efficiency, and each country's contribution to global emissions.
- Overall, the project demonstrated how SQL can be used to extract meaningful insights from real-world data and support data-driven decision-making.

## **Experience/Challenges**

Working on this project provided practical experience in handling real-world datasets using SQL. One of the main challenges was establishing relationships between multiple tables, ensuring consistent column names and datatypes, and writing efficient queries to answer analytical questions. Understanding when to use joins, aggregations, subqueries, and window functions required careful reasoning. Despite these challenges, the project improved my SQL skills and enhanced my ability to analyze and interpret data effectively

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# Questions & Answers

I welcome your questions and feedback.

# THANK YOU